**LECTURE 15**
**BINARY RESPONSE MODELS**

In this lecture, we discuss models in which the dependent variable takes only two values, usually zero or one. For example, $Y_i$ can measure whether or not individual $i$ participates in the labor force: $Y_i = 1$ (yes) or $Y_i = 0$ (no). As we will see, the linear regression model may not be appropriate in such cases.

# Bernoulli trials

The econometrician observes $\{Y_i : i = 1, \ldots, n\}$, where $Y_i$'s are iid random variables. The distribution of $Y_i$ is given by

$$Y_i = \begin{cases} 1 & \text{with probability } \theta, \\ 0 & \text{with probability } 1-\theta. \end{cases}$$

Such a distribution is called Bernoulli($\theta$). We have

$$\begin{aligned} \mathrm{E}[Y_i] &= \theta \\ &= \Pr(Y_i = 1). \end{aligned} \tag{1}$$

The PMF of $Y_i$ can be written as

$$p(y_i, \theta) = \theta^{y_i} (1-\theta)^{1-y_i} \text{ for } y_i = 0, 1.$$

Thus, the log-likelihood function is given by

$$\begin{aligned} \log L_n(\theta) &= \left( n^{-1} \sum_{i=1}^n Y_i \right) \log \theta + \left( 1 - n^{-1} \sum_{i=1}^n Y_i \right) \log(1-\theta) \\ &= \overline{Y}_n \log \theta + \left(1 - \overline{Y}_n\right) \log(1-\theta), \end{aligned}$$

where $\overline{Y}_n = n^{-1} \sum_{i=1}^n Y_i$. It follows that the ML estimator of $\theta$ is the sample mean:

$$\widehat{\theta}_n^{ML} = \overline{Y}_n.$$

Next, we compute

$$\frac{d^2 \log p(Y_i, \theta)}{d\theta^2} = -\frac{Y_i}{\theta^2} - \frac{1 - Y_i}{(1-\theta)^2}.$$

Equation (1) implies that

$$\begin{aligned} \mathrm{E}\left[ \frac{d^2 \log p(Y_i, \theta)}{d\theta^2} \right] &= -\frac{1}{\theta} - \frac{1}{1-\theta} \\ &= -\frac{1}{\theta(1-\theta)}. \end{aligned}$$

Thus, in this model, the information (scalar) is given by

$$I(\theta) = \frac{1}{\theta(1-\theta)},$$

and the results in Lecture 14 imply that

$$\begin{aligned} \overline{Y}_n &\to_p \theta, \\ n^{1/2}\left(\overline{Y}_n - \theta\right) &\to_d N(0, \theta(1-\theta)). \end{aligned}$$

We can estimate the asymptotic variance consistently by $\overline{Y}_n\left(1-\overline{Y}_n\right)$. The $1-\alpha$ asymptotic confidence interval for $\theta$ can be constructed as follows:

$$\left[\overline{Y}_n \pm z_{1-\alpha/2}\sqrt{\frac{\overline{Y}_n\left(1-\overline{Y}_n\right)}{n}}\right].$$

The Bernoulli trials model is a univariate model. Next, we extend it to the case where the probability of $Y_i$ taking on 1 is a function of some exogenous explanatory variables.

# Linear probability model

Suppose that the econometrician observes $\{(Y_i, X_i) : i = 1, \ldots, n\}$, where $Y_i$'s are binary $\{0,1\}$ variables, and $X_i$'s are $k$-vectors of exogenous variables that explain $\Pr(Y_i = 1 \mid X_i)$. Let us assume that

$$\Pr(Y_i = 1 \mid X_i) = F\left(X_i^\top \beta\right),$$

for some function $F$. In contrast to the Bernoulli trials model, the probability of $Y_i$ taking on the value one or zero is not constant and depends on some observable characteristics $X_i$.

If we assume that $Y_i$ and $X_i$ are related by the linear regression model, i.e.,

$$\mathrm{E}[Y_i \mid X_i] = X_i^\top \beta,$$

then

$$F\left(X_i^\top \beta\right) = X_i^\top \beta.$$

In other words, the probability of $Y_i$ taking on 1 is a linear function of $X_i$. Such an assumption is a good starting point for the analysis; however, it suffers from a serious flaw. If some of the $X$'s are continuous, $X_i^\top \beta$ cannot be restricted to the zero-one interval, and the model may produce probabilities that are negative or greater than one. Thus, to avoid nonsense predictions, one has to consider functions $F$ restricted to the zero-one interval, and therefore nonlinear. A natural choice for $F$ is any CDF.

# Probit and logit models

It is convenient to introduce a latent (unobservable) variable $Y_i^*$ for which the usual linear regression model holds:

$$Y_i^* = X_i^\top \beta + U_i.$$

We assume further that $\{(Y_i^*, X_i) : i = 1, \ldots, n\}$ are iid, and that

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* > 0, \\ 0 & \text{if } Y_i^* \leq 0. \end{cases}$$

Let $F$ be the conditional CDF of $U_i$ given $X_i$:

$$F(u \mid X_i) = \Pr(U_i \leq u \mid X_i).$$

Then,

$$\begin{aligned} \Pr(Y_i = 1 \mid X_i) &= \Pr(Y_i^* > 0 \mid X_i) \\ &= \Pr\left(X_i^\top \beta + U_i > 0 \mid X_i\right) \\ &= \Pr\left(U_i > -X_i^\top \beta \mid X_i\right) \\ &= 1 - F\left(-X_i^\top \beta\right). \end{aligned}$$

Assume further that the distribution of $U_i$ does not depend on $X_i$, so we write $F(u)$ instead of $F(u \mid X_i)$. The usual assumption in this framework is that $F$ is symmetric around zero, i.e.,

$$F(u) = 1 - F(-u).$$

Under this assumption, we obtain that

$$\Pr(Y_i = 1 \mid X_i) = F\left(X_i^\top \beta\right).$$

The common choices for $F$ are

- Probit model: $F$ is the standard normal CDF, denoted by $\Phi$.
- Logit model: $F$ is the logistic CDF, denoted by $\Lambda$:

$$\begin{aligned} \Lambda\left(X_i^\top \beta\right) &= \frac{\exp\left(X_i^\top \beta\right)}{\exp\left(X_i^\top \beta\right) + 1} \\ &= \frac{1}{1 + \exp\left(-X_i^\top \beta\right)}. \end{aligned}$$

For either choice of $F$, the model is estimated by the ML method. The PMF of $Y_i$ conditional on $X_i = x_i$ is similar to the PMF of $Y_i$ in the Bernoulli trials model, but with $\theta$ replaced by $F\left(x_i^\top \beta\right)$:

$$p(y_i, \beta \mid x_i) = F\left(x_i^\top \beta\right)^{y_i} \left(1 - F\left(x_i^\top \beta\right)\right)^{1-y_i} \text{ for } y_i = 0, 1.$$

Thus, the log-likelihood is given by

$$\log L_n(\beta) = n^{-1} \sum_{i=1}^{n} Y_i \log F\left(X_i^\top \beta\right) + n^{-1} \sum_{i=1}^{n} (1 - Y_i) \log\left(1 - F\left(X_i^\top \beta\right)\right).$$

The first-order condition for $\widehat{\beta}_n^{ML}$ is

$$\begin{aligned} 0 &= \frac{\partial \log L_n\left(\widehat{\beta}_n^{ML}\right)}{\partial \beta} \\ &= n^{-1} \sum_{i=1}^{n} Y_i \frac{\partial F\left(X_i^\top \widehat{\beta}_n^{ML}\right) / \partial \beta}{F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} - n^{-1} \sum_{i=1}^{n} (1 - Y_i) \frac{\partial F\left(X_i^\top \widehat{\beta}_n^{ML}\right) / \partial \beta}{1 - F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} \\ &= n^{-1} \sum_{i=1}^{n} Y_i \frac{f\left(X_i^\top \widehat{\beta}_n^{ML}\right)}{F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} X_i - n^{-1} \sum_{i=1}^{n} (1 - Y_i) \frac{f\left(X_i^\top \widehat{\beta}_n^{ML}\right)}{1 - F\left(X_i^\top \widehat{\beta}_n^{ML}\right)} X_i, \end{aligned}$$

where $f$ is the density corresponding to $F$:

$$f(u) = \frac{dF(u)}{du}.$$

There is no closed-form expression for $\widehat{\beta}_n^{ML}$, and it must be computed numerically. Statistical software packages can produce the ML estimates and their asymptotic standard errors for probit and logit models, given the data on $Y_i$ and $X_i$.

In the logit model, the density of $\Lambda$ satisfies

$$\frac{d\Lambda(u)}{du} = \Lambda(u)\left(1 - \Lambda(u)\right). \tag{2}$$

Therefore, for the logit model,

$$
\begin{aligned}
\frac{f\left(X_i^\top \beta\right)}{F\left(X_i^\top \beta\right)} &= 1-\Lambda\left(X_i^\top \beta\right), \\
\frac{f\left(X_i^\top \beta\right)}{1-F\left(X_i^\top \beta\right)} &= \Lambda\left(X_i^\top \beta\right).
\end{aligned}
$$

The first-order condition simplifies to

$$
\begin{aligned}
\frac{\partial \log L_n\left(\widehat{\beta}_n^{ML}\right)}{\partial \beta} &= n^{-1}\sum_{i=1}^n \left(Y_i-\Lambda\left(X_i^\top \widehat{\beta}_n^{ML}\right)\right) X_i \\
&= 0.
\end{aligned}
$$

Similarly, we have that

$$
\frac{\partial \log p\left(y_i,\beta \mid x_i\right)}{\partial \beta} = \left(y_i-\Lambda\left(x_i^\top \beta\right)\right) x_i,
$$

and

$$
\frac{\partial^2 \log p\left(y_i,\beta \mid x_i\right)}{\partial \beta \partial \beta^\top} = -\Lambda\left(x_i^\top \beta\right)\left(1-\Lambda\left(x_i^\top \beta\right)\right) x_i x_i^\top.
$$

Thus, the information matrix is given by

$$
\begin{aligned}
I(\beta) &= -\mathrm{E}\left[\frac{\partial^2 \log p\left(Y_i,\beta \mid X_i\right)}{\partial \beta \partial \beta^\top}\right] \\
&= \mathrm{E}\left[\Lambda\left(X_i^\top \beta\right)\left(1-\Lambda\left(X_i^\top \beta\right)\right) X_i X_i^\top\right].
\end{aligned}
$$

Hence, for the logit model, $\widehat{\beta}_n^{ML} \to_p \beta$, and

$$
n^{1/2}\left(\widehat{\beta}_n^{ML}-\beta\right) \to_d N\left(0, \left(\mathrm{E}\left[\Lambda\left(X_i^\top \beta\right)\left(1-\Lambda\left(X_i^\top \beta\right)\right) X_i X_i^\top\right]\right)^{-1}\right).
$$

The asymptotic variance-covariance matrix of $\widehat{\beta}_n^{ML}$ can be estimated consistently by

$$
\left(n^{-1}\sum_{i=1}^n \Lambda\left(X_i^\top \widehat{\beta}_n^{ML}\right)\left(1-\Lambda\left(X_i^\top \widehat{\beta}_n^{ML}\right)\right) X_i X_i^\top\right)^{-1}.
$$

## Marginal effects

In the linear regression model, the marginal effect of $X_i$ on $\mathrm{E}\left[Y_i \mid X_i\right]$ is $\beta$. In nonlinear binary choice models such as probit or logit,

$$
\begin{aligned}
\frac{\partial \mathrm{E}\left[Y_i \mid X_i\right]}{\partial X_i} &= \frac{\partial \Pr\left(Y_i=1 \mid X_i\right)}{\partial X_i} \\
&= \frac{\partial F\left(X_i^\top \beta\right)}{\partial X_i} \\
&= f\left(X_i^\top \beta\right)\beta,
\end{aligned}
$$

and

$$
\begin{aligned}
\frac{\partial \Pr\left(Y_i=0 \mid X_i\right)}{\partial X_i} &= \frac{\partial\left(1-\Pr\left(Y_i=1 \mid X_i\right)\right)}{\partial X_i} \\
&= -f\left(X_i^\top \beta\right)\beta.
\end{aligned}
$$

Thus, in the case of the probit model, the marginal effect of $X_i$ on $\Pr(Y_i = 1 \mid X_i)$ is given by

$$\phi\left(X_i^\top \beta\right)\beta,$$

where $\phi(u) = d\Phi(u)/du$ is the standard normal density. In the case of logit, equation (2) implies that $\partial \Pr(Y_i = 1 \mid X_i)/\partial X_i$ is given by

$$\Lambda(X_i^\top \beta)\left(1 - \Lambda(X_i^\top \beta)\right)\beta.$$

The marginal effects can be estimated by replacing the unknown coefficients $\beta$ with their ML estimators. Fix $X_i = x$. Then

$$\frac{\partial \widehat{\Pr(Y_i = 1 \mid X_i = x)}}{\partial X_i} = f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML}.$$

By Slutsky's theorem and consistency of $\widehat{\beta}_n^{ML}$,

$$\begin{aligned}
\frac{\partial \widehat{\Pr(Y_i = 1 \mid X_i)}}{\partial X_i} &= f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML} \\
&\to_p f\left(x^\top \beta\right)\beta \\
&= \frac{\partial \Pr(Y_i = 1 \mid x)}{\partial X_i},
\end{aligned}$$

provided that $f$ is continuous.

The asymptotic distribution of $f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML}$ can be obtained using the delta method:

$$\begin{aligned}
n^{1/2}\left(f\left(x^\top \widehat{\beta}_n^{ML}\right)\widehat{\beta}_n^{ML} - f\left(x^\top \beta\right)\beta\right) &\to_d N\left(0, \frac{\partial\left(f\left(x^\top \beta\right)\beta\right)}{\partial \beta^\top} I^{-1}(\beta)\frac{\partial\left(f\left(x^\top \beta\right)\beta^\top\right)}{\partial \beta}\right) \\
&= N\left(0, \left(f'\left(x^\top \beta\right)\beta x^\top + f\left(x^\top \beta\right)I_k\right)I^{-1}(\beta)\left(f'\left(x^\top \beta\right)\beta x^\top + f\left(x^\top \beta\right)I_k\right)^\top\right).
\end{aligned}$$